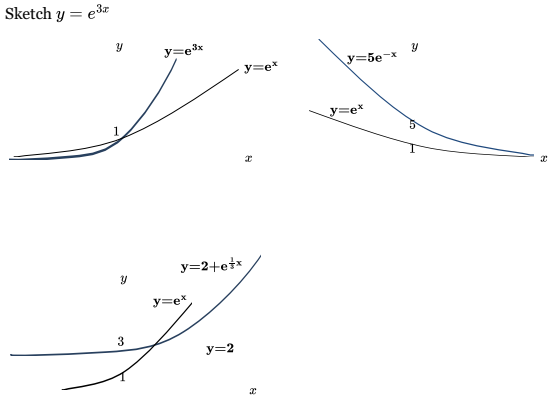
1.25 Graph Sketching: e and Natural Logarithms

Chapter 14 · Original slide 11 · More Graph Transformations



Sketch $y = 5e^{-x}$

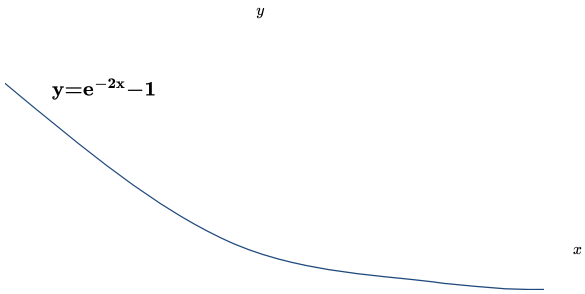
Note: Recall the shape of a $y = a^{-x}$ graph.

5 is 'outside' function, so stretch of scale factor 5 on y-axis.

Important Note: Because the original asymptote was $y = 0$, it is now $y = 2$ and you must indicate this along with its equation.

Sketch $y = 2 + e^{\frac{1}{3}x}$

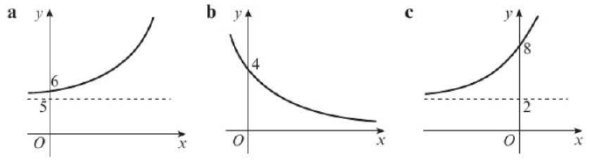
Chapter 14 · Original slide 12 · Test Your Understanding



Sketch $y = e^{-2x} - 1$

1.25 Graph Sketching: e and Natural Logarithms

Chapter 14 · Original slide 13 · Exercise 14B (Pages 316-317)



4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.

Chapter 14 · Original slide 14 · Exercise 14B (Pages 316-317)

6 Differentiate the following with respect to x.

a e^{6x} **b** $e^{-x/3}$ **c** $7e^{2x}$ **d** $5e^{0.4x}$ **e** $e^{3x} + 2e^{x}$ **f** $e^{x}(e^{x} + 1)$

1.25 Graph Sketching: e and Natural Logarithms

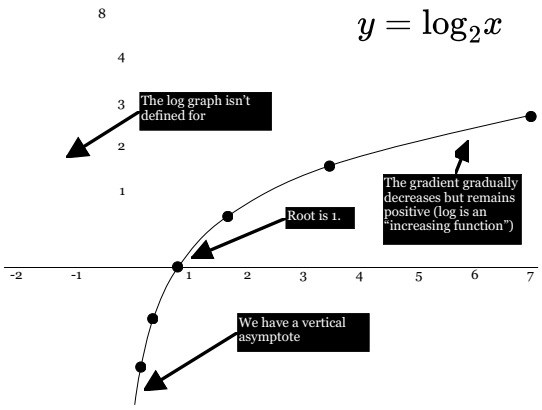
Chapter 14 · Original slide 15 · Exercise 14B (Pages 316-317)

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

8 The function f is defined as $f(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

Chapter 14 · Original slide 27 · Exponentials and logarithms



x	0.25	0.5	1	2	4	8
y						

1.25 Graph Sketching: e and Natural Logarithms

Chapter 14 · Original slide 41 · Natural Logarithms

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.

We also say that e^x is "the" exponential function.

The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

! The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Since "e to the power of" and "ln of" are inverse functions, they cancel each other out!

Solve $e^x = 5$

Solve $2 \ln x + 1 = 5$

"ln both sides".

On the LHS it cancels out the "e to the power of"

Do "e to the power of" each side. On the LHS it cancels out the ln.